

The curtain opener circuit comprises a power supply and a control section. The power supply section is built around transformer X1, bridge rectifier comprising diodes D1 through D4, filter capacitors and regulator 78S06. The 50Hz, 230V AC mains is stepped down by transformer X1 to deliver a secondary output of 9V, 1A. The transformer output is rectified by a bridge rectifier, filtered by capacitor C1 and regulated by IC 78S06 (IC1). Capacitor C2 bypasses the ripples present in the regulated supply.

The control section is built around a dual D-type flip-flop CD4013 (IC2), motor driver L293D (IC3), two push-to-on switches (S1 and S3), two reed switches (S2 and S4) and a few discrete

components. A geared motor (M1) is used to drive the mechanical pulley assembly for opening and closing the curtain.

To open the curtain, press switch S1 momentarily. Geared motor M1 starts running in clockwise direction. When the curtain is completely opened, a small magnet attached to the curtain comes close to reed switch S2 fitted to the door/window frame and the reed switch closes. This resets IC2(A) and the motor stops.

Similarly, to close the curtain, press switch S3 momentarily. Geared motor M1 starts running in counter-clockwise direction. When the curtain is completely closed, the magnet comes close to reed switch S4 fitted to the door/window frame

on the opposite side. The reed switch closes to reset IC2(B) and the motor stops.

Assemble the circuit on a general-purpose PCB and house in a small case. Install the geared motor on the top centre of the door and place reed switches at the top left and right frames of the door. Connect switches S1 and S3 to the circuit using a four-wire cable. Hang these switches to either left or right side of the door for operating the motor to open or close the curtain.

EFY note. At EFY, this circuit was tested without mechanical assembly. You can design the mechanical assembly taking into account the dimensions of the doors or windows. ●

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